



AIM-340
AIM-341
AIM-342
AIR ENTRAINMENT
METER



Aimil Ltd.

Instrumentation & Technologies



Air Entrainment Meter

The equipment comprises of the following :

1. Air Entrainment Meter Main Assembly consisting of :

i. Measuring Bowl	One 5 dm ³ capacity	One 10 dm ³ capacity	One 100 dm ³ capacity
ii. Conical Cover and Assembly complete with one stand pipe	One to fit on the bowl	One to fit on the bowl	One to fit on the bowl
iii. Spray Tube with funnel	1 No.	1 No.	1 No.
2. Air Pressure Bulb	1 No.	1 No.	1 No.
3. Foot Pump	----	—	1 No.
4. Calibration Cylinder with spring leaf for holding the cylinder	1 No.	1 No.	1 No.
5. Measure	1 No.	1 No.	1 No.
6. Rubber Mallet	1 No.	1 No.	1 No.
7. Strike-off Bar	1 No.	1 No.	1 No.
8. Trowel	1 No.	1 No.	1 No.
9. Tamping Rod	1 No.	1 No.	1 No.
10. Copy of Operating Instructions with General Assembly Drawing.			1 No.



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Operating Instructions

Air Entrainment Meter

Introduction

The equipment meets the requirements of IS : 1199, IS : 9799 and ASTM C 231 for determining the air content of freshly mixed concrete by the pressure method. This method is considered adequate for all ordinary types of concrete and mortar, except for concrete or mortar made with highly porous aggregates, where the aggregate correction factor cannot be determined accurately by the technique found satisfactory for usual types of relatively dense natural aggregate.

Three models of Air Entrainment Meters are manufactured :

- | | |
|------------|---|
| 1. AIM-340 | Air Entrainment Meter 5 dm ³ capacity for maximum size of aggregate 38mm. |
| 2. AIM-341 | Air Entrainment Meter 10 dm ³ capacity for maximum size of aggregate 75mm. |
| 3. AIM-342 | Air Entrainment Meter 100 dm ³ capacity for maximum size of aggregate 150mm. |

Description

The equipment is illustrated in the General Assembly Drawing and the numbers given against the components in the description below pertain to this General Assembly Drawing.

AIM-340

It consists of a measuring bowl (3), a conical cover assembly (4), a measure (2), a spray tube with funnel (10), a rubber mallet (11) and a strike-off bar.



a. Measuring Bowl

The flanged metallic bowl is not readily attacked by the cement paste having a diameter equal to about 1 to 1.25 times the height. The outer rim and upper surface of the flange as well as the interior surfaces of the bowl are smooth. The bowl is pressure tight and sufficiently rigid to limit the expansion factor 'D' of the apparatus.

b. Conical Cover Assembly (4)

The flanged metallic cover is not readily attacked by the cement paste, as interior surfaces are inclined from the horizontal. The outer rim and lower surface of the flange and the sloping interior surface are pressure tight and sufficiently rigid to limit the expansion factor of the assembly. The cover is fitted with a stand pipe (7). The graduations are for the range of 0 - 10% of air content. The internal diameter of the stand pipe (7) is so designed that under the normal operating pressure the water column is lowered sufficiently to measure air contents upto 0.1 percent. The applied air pressure is indicated by a pressure gauge (8) connected to the air chamber above the water column on the stand pipe (7). The cover is fitted with air valves (6) for venting air at the top and a petcock (13) to fill water as required. Clamps (5) are provided for clamping the cover to the bowl to make a pressure tight seal without entrapping air at the joint between the flanges of the cover and bowl. A pressure bulb (14) is provided for applying the required pressure to the specimen. A base (1) is provided for keeping the whole assembly in the vertical position.

c. Calibration Cylinder (9)

It consists of a cylindrical metallic measure with slots at the lower end.

d. Spring (10)

A Spring leaf is provided for holding the calibration cylinder in place.

e. Spray Tube with Funnel (12)

A tube of appropriate diameter with funnel so constructed that when water is added there is minimum disturbance to the concrete in the measuring bowl.



f. Rubber Mallet (11)

It is used for tapping sides of the bowl and conical cover.

g. Measure (2)

For filling the indicator stand pipe with water from the top of the concrete to the zero mark.

h. Strike-off Bar

A strike-off bar is provided for levelling the concrete.

i. Trowel

j. Tamping Rod

It is used to consolidate each layer of concrete by rodding.

AIM-341

It is similar to AIM-340 Air Entrainment Meter, but its capacity is 10 dm³ and suitable for aggregate with maximum size of 75 mm.

AIM-342

It is also similar to AIM-341 Air Entrainment Meter, but its capacity is 100 dm³ and is suitable for aggregates of a maximum size of 150mm. A foot pump in place of pressure bulb and a holding spring for calibrating cylinder are provided.

PROCEDURE

The procedure for all three models of the Air Entrainment Meter is the same and is described below :

A change in barometric pressure is caused by changes in altitude or by changes of temperature and humidity. It should be noted that rough handling will effect the calibration of pressure type apparatuses for the determination of air content.



The following procedure should be adopted to check the operating pressure.

Calibration of Calibration Cylinder

The weight of water 'w' (in grams) required to fill the calibration cylinder shall be accurately determined by using a scale sensitive to 0.5g.

Calibration of Measuring Bowl

The weight of water 'W' (in grams) required to fill the measuring bowl shall be determined, using a scale sensitive to 0.1% of the weight of the bowl filled with water.

A glass plate is placed carefully over the flange of the bowl in such a manner as to ensure that the bowl is completely filled with water. A thin film of cup grease smeared on the flange of the bowl will make a water tight joint between the glass plate and the top of the bowl.

Determination of Constant 'R'

The constant 'R' represents the volume of the calibration cylinder (w) expressed as a percentage of the volume of the measuring bowl (W). 'R' is calculated as follows :

$$R = \frac{100w}{W}$$

Determination of Expansion Factor 'D'

The expansion factor 'D' for the assembly shall be determined by filling the apparatus with only water (making certain that all entrapped air has been removed and the water level is exactly on the zero mark) and applying an air pressure approximately equal to the operating pressure P, as determined by the calibration test. The amount by which the water column is lowered shall be the equivalent expansion factor 'D' for that particular apparatus and pressure.

Determination of Calibration Factor 'K'

The calibration factor 'K' is the amount by which the water column shall be depressed, during the calibration procedure to obtain the gauge pressure required to make the graduations on the glass tube correspond directly to the percentage of air introduced into the measuring bowl by the calibration cylinder when the bowl is level full of water. The calibration factor 'K' is computed as follows :

$$K = 0.98 R + D \quad \dots (ii)$$

***Calibration Test to Determine Operating Pressure 'p' on Pressure Gauge***

Invert the calibration cylinder and place it at the centre of the dry bottom of the measuring bowl and carefully lower the conical cover over the measuring bowl with seal. After the cover is clamped in place, carefully adjust the apparatus assembly to the vertical position and add water at room temperature by means of the tube and funnel, until it rises above the zero mark on the stand pipe. Close the vent and pump air into the apparatus to the approximate operating pressure.

Incline the assembly to about 30° from the vertical and using the bottom of the bowl as a pivot, describe several complete circles with the upper end of stand pipe while simultaneously tapping the cover and sides of the bowl lightly to remove any entrapped air in the inner surfaces of the apparatus. Then open the vent. Bring the water level exactly to the zero mark by bleeding water through the petcock in the top of the conical cover. After closing the vent apply pressure until the water level has dropped to an amount equivalent to about 0.1 percent of air more than the value of the calibration factor K. To relieve local restraints lightly tap the sides of the bowl. When the level is exactly at the value of the calibration factor 'K', read the pressure 'P' indicated by the gauge and record it to the nearest 0.01 kgf/cm^2 .

Gradually release the pressure and open the vent to determine whether the water level returns to the zero mark, while lightly tapping the sides of the bowl. Failure of the water level to return indicates loss of air from the calibration cylinder or loss of water due to a leak in the assembly. If the water level fails to return to within 0.05% of air content of the zero mark and no leakage beyond few drops of water is found, some air was probably lost from the calibration cylinder. If the leakage is more than a few drops of water, tighten the leakage joint before repeating the calibration procedure. Check the indicated pressure reading promptly by bringing the water level exactly to the zero mark, closing the vent and applying the pressure p, just determined. Tap the gauge lightly with a finger. Then the gauge indicates the exact pressure p. The water column should read the value of the calibration factor K, used in the first pressure application within about 0.05% of air.

***Determination of Aggregate Correction Factor 'G'***

The aggregate correction factor shall be determined in a combined sample of fine and coarse aggregates.

The weights of fine and coarse aggregates present in the volume 'S' of the sample of fresh concrete whose air content is to be determined, shall be determined as follows :

$$F_s = \frac{S}{B} \times F_b$$

$$C_s = \frac{S}{B} \times C_b$$

Where

F_s = weight in kg of fine aggregate in concrete sample under test

S = volume in dm^3 of concrete sample (same as the volume of the measuring bowl of apparatus)

B = volume in dm^3 of concrete produced per batch

F_b = total weight in kg of fine aggregate in batch

C_s = weight in kg of coarse aggregate in concrete sample under test

C_b = total weight in kg of coarse aggregate in batch.

Mix a representative sample of fine aggregate of weight F_s and coarse aggregate of weight C_s and place its in the measuring bowl filled one-third full of water. Add the mixed aggregate in small amounts at a time, until all the aggregate is inundated. Add each scoopful in a manner that will entrap as little air as possible and remove accumulations of foam promptly. Tap the sides of the bowl and lightly rod the upper layer of the aggregate about ten times and stir after each addition of fine aggregate to eliminate entrapped air.



When all of the aggregate has been placed in the bowl and inundated for at least 5 minutes, strike-off all foam and excess water and thoroughly clean the flanges of both the bowl and conical cover so that when the cover is clamped in place, a pressure tight seal is obtained. Complete the test as described below. The aggregate correction factor $G = h_1 - h_2$ as determined in the tests on the aggregate, h_1 = reading at pressure p and h_2 = reading at zero pressure after release of pressure p .

PROCEDURE FOR DETERMINING AIR CONTENT OF CONCRETE

Place a representative sample of the concrete in the measuring bowl in three equal layers, consolidating each layer by rodding the bowl. vibration may be substituted for rodding and by tapping the sample when the air content of concrete placed by vibration is to be determined. When the concrete is to be placed by rodding, consolidate each layer of concrete by about 25 strokes of the tamping rod evenly distributed over the cross-section. Follow the rodding of each layer by tapping the sides of the bowl smartly 10 to 15 times with the mallet until the cavities left by rodding are levelled out and no large bubbles of the air appear over the surface of the rodded layer. In rodding the first layer, the rod shall not forcibly strike the bottom of the bowl. In rodding the second and final layer, only enough force shall be used to cause the rod to penetrate the surface of the previous layer. Slightly overfill the bowl with the third layer and after rodding or vibration remove the excess concrete by sliding the strike-off bar across the top flange with a sawing motion until the bowl is just full to the flange level. Thoroughly clean the flanges of the bowl and of the conical cover so that when the cover is clamped in place, a pressure-tight seal will be obtained. Assemble the apparatus and add water over the concrete by means of the tube until it rises to about the half way mark in the stand pipe. Incline the apparatus assembly about 30° from vertical end using the bottom of the bowl as a pivot, describe several complete circles, with the upper end of the column simultaneously tapping the conical cover lightly to remove any entrapped air bubbles above the concrete sample. Replace the apparatus assembly to its vertical position and fill the water column slightly above the zero mark, while lightly tapping the sides of the bowl. Foam on the surface of the water column may be removed with a syringe or with a spray of alcohol to provide a clear meniscus.

Bring the water level to the zero mark of the stand pipe, closing the vent at the top of the water column. Apply slightly more than the desired test pressure p (0.02 kg/cm^2 more) to the concrete by means of a hand pump. To relieve local restraints, tap the sides of the measures smartly, and when the pressure gauge indicates the exact test pressure p (as determined in accordance with the calibration test) read the level h_1 and record the value to the nearest division or half division (0.10 to 0.05% air content) on the graduated stand pipe.



For extremely harsh mixes, it may be necessary to tap the bowl vigorously until further tapping produces no change in the indicated air content. Gradually release the air pressure through the vent at the top of the water column and tap the sides of the bowl lightly for about one minute.

Record the water level h_2 to the nearest division or half division. The apparent air content A_1 is equal to $h_1 - h_2$. Repeat the step specified as above (without adding water to re-establish the water level at the zero mark). The two consecutive determinations of apparent air content should check within 0.2% of air and shall be averaged to give the value A_1 , to be used in calculating the air content, A as given below :

Calculations

Calculate the air content of the concrete as follows :

$$A = A_1 - G$$

Where

- A = air content, percentage by volume of concrete.
 A_1 = apparent air content, percentage by volume of concrete.
 G = aggregate correction factor, percentage by volume of concrete.

Precautions

1. The operating pressure p should be checked periodically as per the calibration procedure.
2. During the calibration procedure the apparatus assembly should not be moved from the vertical position until pressure has been applied which will force water about one-third of the way up into the calibration cylinder. Any loss of air from the cylinder will nullify the calibration.
3. The aggregate correction factor will vary with different aggregates and has to be verified each time.

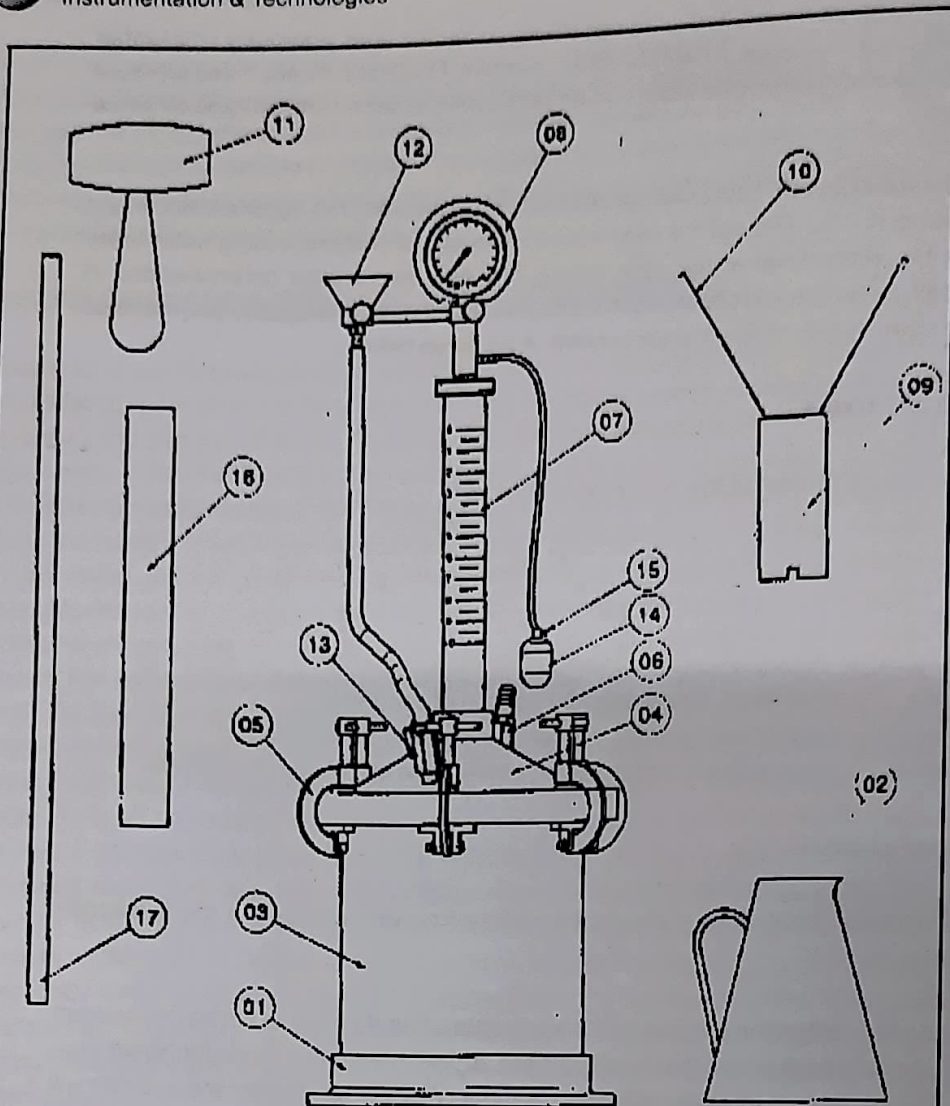
Maintenance

The apparatus should always be kept clean.



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AIM-340, 341, 342 AIR ENTRAINMENT METER

S.NO	DESCRIPTION
01	BASE
02	MEASURE
03	MEASURING BOWL
04	CONICAL COVER ASSEMBLY
05	CLAMPS
06	AIR VALVE
07	STAND PIPE
08	PRESSURE GAUGE
09	CALIBRATION CYLINDER

S.NO	DESCRIPTION
10	SPRING
11	MALLET
12	SPRAY TUBE WITH FUNNEL
13	BALL VALVE
14	PRESSURE BULB
15	RELEASE VALVE
16	STRAIGHT EDGE BAR
17	TEMPING ROD
18	